

SmartStock AI

Inventory Management System

Internal Policies & Procedures

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1. Stock Management Policy

1.1 Purpose

This policy establishes guidelines for managing inventory levels, stock rotation, and warehouse operations at SmartStock AI. Effective stock management ensures product availability while minimizing carrying costs and waste.

1.2 Stock Level Categories

All inventory items are classified into four stock status levels based on quantity available relative to reorder points:

Status	Definition	Action Required
Critical	Below 50% of reorder point	Immediate reorder + management alert
Low	Below reorder point	Auto-reorder triggered
Adequate	Between reorder point and 2x reorder	Monitor weekly
Surplus	Above 2x reorder point	Review for markdown or transfer

1.3 Safety Stock Requirements

Minimum safety stock levels by product category (expressed in days of supply):

Category	Safety Stock (Days)	Max Shelf Life
Electronics	7	N/A
Groceries & Perishables	3	7 days
Clothing & Apparel	14	N/A
Home & Garden	10	N/A
Health & Beauty	7	90 days

1.4 Cycle Counting

Physical inventory counts are conducted on a rotating schedule:

- A-class items (top 20% by value): Monthly counts
- B-class items (middle 60% by value): Quarterly counts
- C-class items (bottom 20% by value): Semi-annual counts

All discrepancies exceeding 2% must be investigated within 48 hours and reported to the Operations Manager.

1.5 Shrinkage Control

The shrinkage target is 1.2% of total inventory value. Current shrinkage rate as of Q2 2026 is 0.9%. Automatic audit triggers when shrinkage exceeds 2% for any category. All shrinkage events above \$500 require incident reports filed within 24 hours.

2. Purchase Order Policy

2.1 PO Creation Authority

Purchase orders may be created by the following roles:

- AI Auto-Order System: Automatically generates POs for items hitting reorder points under Rule Set A (top 20% velocity). Auto-approval limit: \$15,000 per PO.
- Manager: Can create and approve POs up to \$25,000.
- Admin: Can create and approve POs of any value.

2.2 Approval Workflow

All purchase orders follow this approval chain:

- Step 1: PO created (status: draft)
- Step 2: Submitted for approval (status: pending_approval)
- Step 3: Manager/Admin reviews and approves (status: approved)
- Step 4: PO sent to supplier (status: sent)
- Step 5: Supplier confirms delivery (status: confirmed)

POs above \$10,000 require two-factor approval (two separate managers or one admin).

2.3 Reorder Rules

SmartStock AI uses three automated reorder rule sets:

Rule Set	Applies To	Reorder Trigger	Auto-Approve Limit
A (Fast)	Top 20% velocity	14-day demand threshold	\$15,000
B (Medium)	Middle 60% velocity	21-day demand threshold	\$10,000
C (Slow)	Bottom 20% velocity	45-day demand threshold	\$5,000 (manual)

2.4 PO Cancellation

Purchase orders may be cancelled only under the following conditions:

- Before status changes to "sent": Free cancellation with reason code.
- After "sent" but before delivery: Cancellation fee of 5% of PO value may apply.
- After delivery: Must follow Returns & Refunds Policy (Section 5).

3. Supplier Management Policy

3.1 Supplier Qualification

All new suppliers must complete the following qualification process before being added to the approved vendor list:

- Step 1: Initial assessment by Procurement team (5 business days)
- Step 2: Financial stability check and credit reference verification
- Step 3: Quality audit or site visit for critical suppliers
- Step 4: Trial order with quality inspection upon receipt
- Step 5: Final approval by Procurement Manager and Quality Assurance

3.2 Supplier Performance Scoring

Supplier reliability is measured on a 100-point scale based on:

Metric	Weight	Target
On-Time Delivery	40%	$\geq 95\%$
Order Accuracy	25%	$\geq 98\%$
Quality Pass Rate	20%	$\geq 99\%$
Response Time	15%	≤ 24 hours

3.3 Supplier Status Levels

- Preferred (Score $\geq 96\%$): Priority ordering, extended payment terms, exclusive partnerships.
- Approved (Score 90-95%): Standard ordering, normal payment terms.
- Probationary (Score 80-89%): Increased oversight, weekly performance reviews, improvement plan required within 30 days.
- Suspended (Score $< 80\%$): No new orders until score improves. Dual-sourcing strategy activated.

3.4 Supplier Diversity

SmartStock AI maintains a minimum of 2 qualified suppliers for any single product category to mitigate supply chain risk. No single supplier may represent more than 40% of total procurement spend.

4. Invoice Processing Policy

4.1 Invoice Submission

All supplier invoices must be submitted through the SmartStock AI invoice scan system or manually entered into the platform. Invoices received via email or paper are not considered submitted until digitized.

4.2 Required Invoice Data

Every invoice must contain the following fields for processing:

- Supplier name (must match approved vendor list)
- Invoice number (unique per supplier)
- Invoice date and due date
- Line items with: product name, SKU code, quantity, unit price
- Total amount and tax breakdown
- Currency code (USD for domestic, ISO 4217 for international)

4.3 Invoice Verification

Invoices go through a three-way match process:

- Match 1: Invoice line items against Purchase Order quantities and prices
- Match 2: Goods receipt against Purchase Order quantities delivered
- Match 3: Supplier statement against invoice amounts

Discrepancies exceeding 5% or \$500 (whichever is lower) require manual review by the Finance team before approval.

4.4 Payment Terms

Standard payment terms by supplier tier:

- Domestic suppliers: Net 30 days
- International suppliers: Net 45 days
- Early payment discount: 2/10 Net 30 (2% discount if paid within 10 days)
- Preferred suppliers: Net 45 days (extended terms)

4.5 Late Payment Penalties

Payments received after the due date incur a 1.5% monthly late fee. Suppliers with 3 or more late payments in a rolling 12-month period are placed on probationary status and must be pre-approved for future orders.

5. Returns & Refunds Policy

5.1 Return Windows by Category

Return eligibility is determined by product category:

Category	Return Window	Condition Required	Restocking Fee
Electronics	30 days	Original packaging	15%
Groceries	24 hours	Receipt + product in hand	None
Clothing	60 days	Tags attached, unworn	Free
Home & Garden	45 days	Unused, original packaging	10%
Health & Beauty	30 days	Sealed only	None

5.2 Return Process

- Step 1: Customer initiates return via portal or contacts support
- Step 2: Return Merchandise Authorization (RMA) number generated
- Step 3: Product shipped back or returned in-store
- Step 4: Quality inspection within 3 business days
- Step 5: Refund processed within 5 business days of inspection

5.3 Non-Returnable Items

The following items cannot be returned:

- Custom or personalized orders
- Hazardous materials or chemicals
- Perishable goods past the 24-hour window
- Items with removed or damaged serial numbers
- Digital products or software licenses after activation

5.4 Defective Product Handling

Products reported as defective within the first 7 days are eligible for immediate replacement at no cost. After 7 days but within the return window, defective products are repaired or replaced at SmartStock AI discretion. All defective items are tagged in the system for quality tracking and supplier notification.

6. AI & Automation Policy

6.1 AI Decision Authority

SmartStock AI uses artificial intelligence for demand forecasting, reorder automation, and anomaly detection. AI systems operate within the following boundaries:

- Forecasting: AI generates demand predictions with MAPE targets below 6%. Forecasts are advisory; human approval is required for orders above \$10,000.
- Auto-Reorder: AI can automatically generate purchase orders under Rule Set A without human approval, up to \$15,000 per PO. All auto-generated POs are logged and auditable.
- Anomaly Detection: AI flags unusual inventory movements, pricing discrepancies, and supplier anomalies for human review.

6.2 Human Override

All AI-generated recommendations can be overridden by authorized users (Manager or Admin). Overrides are logged with the reason code and the overriding user identity. AI systems learn from overrides to improve future recommendations.

6.3 AI Model Performance

AI model performance is reviewed monthly by the AI & Data Science team:

- Demand forecasting MAPE must remain below 8% per category
- Anomaly detection false positive rate must remain below 15%
- Auto-reorder accuracy must achieve 90% satisfaction rate
- Models are retrained quarterly with the latest sales data

6.4 Invoice Scanning

AI-powered invoice scanning extracts data from uploaded invoices. Extracted data must be verified by a human before confirming and adding to inventory. The system supports PDF, JPEG, and PNG formats. Maximum file size is 5MB. The AI confidence score for each field is displayed for review.

6.5 Chat Assistant

The AI chat assistant operates in two modes:

- Search Stock (NL Query): Queries the live inventory database for stock levels, sales data, forecasts, and supplier information. Responses are grounded in real-time data.
- Search Documents (RAG): Searches uploaded policy documents, reports, and manuals. Responses include source citations with page numbers.

The chat assistant does not have access to modify inventory data. It is read-only by design.

7. Data Security & Privacy Policy

7.1 Data Classification

All data within SmartStock AI is classified into three levels:

- Confidential: API keys, database credentials, encryption keys, user passwords. Never stored in logs or plain text.
- Internal: Inventory data, financial records, supplier contracts, employee information. Access restricted to authorized personnel.
- Public: Product catalog names, general company information. May be shared externally.

7.2 Access Control

Role-based access control (RBAC) is enforced across all system components:

Role	Inventory	Purchasing	Reports	Settings
Viewer	Read	Read	Read	None
Manager	Read/Write	Read/Write/Approve	Read/Write	Limited
Admin	Full	Full	Full	Full

7.3 Data Retention

- Transaction records: 7 years minimum retention
- Audit logs: 5 years minimum retention
- AI training data: Anonymized and retained indefinitely
- User accounts: Retained while active, deactivated accounts archived after 90 days
- Deleted inventory records: Soft-deleted, recoverable for 30 days

7.4 API Security

All API communications are secured with the following measures:

- HTTPS/TLS 1.3 encryption for all data in transit
- JWT authentication with short-lived access tokens (15 minutes) and refresh tokens (7 days)
- Rate limiting: 100 requests per minute per user for standard endpoints, 20 requests per minute for AI endpoints
- CORS restricted to approved domains only
- Quarterly penetration testing by external security firm

7.5 Incident Response

In the event of a security incident:

- Step 1: Incident reported to Security team within 1 hour of detection
- Step 2: Impact assessment completed within 4 hours
- Step 3: Affected systems isolated if necessary
- Step 4: Affected parties notified within 72 hours per GDPR requirements
- Step 5: Post-incident review and remediation plan within 14 days